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Whole class session:

Back pain and anaemia in elderly: a patient with multiple myeloma

Dr YY Hwang

Mr K



- M/74
 - Benign prostatic hypertrophy
 - Hypertension on antihypertensives
 - Diabetes on oral hypoglycaemic agents
-
- Comes in to medical ward with three months history of reduced exercise tolerance and low back pain



- M/74
- BPH
- HT
- DM

CBC

This admission

WBC	12.77 H	9.33	7.33	7.35	9.00	3.89 - 9.93
RBC	2.83 L	2.44 L	2.40 L	2.24 L	2.20 L	3.87 - 5.11
HGB	9.3 L	7.7 L	7.6 L	7.3 L	7.1 L	11.5 - 14.8
HCT	0.271 L	0.233 L	0.225 L	0.209 L	0.205 L	0.347 - 0.449
MCV	95.8 H	95.5	93.8	93.3	93.2	82.0 - 95.5
MCH	32.9 H	31.6	31.7	32.6 H	32.3	27.0 - 32.4
MCHC	34.3	33.0	33.8	34.9	34.6	32.1 - 34.9
RDW	14.7 H	14.7 H	14.7 H	14.6	14.9 H	11.8 - 14.6
PLT	164 L	131 L	131 L	124 L	131 L	167 - 396

WBC DIFFERENTIAL

DC Type	MACHINE	MACHINE	MACHINE	MACHINE	MACHINE	
#Neutrophil	11.50 H	7.56 H	5.82	5.74	7.34	2.01 - 7.42
#Lymphocyte	0.63 L	1.09	0.91 L	1.00 L	1.32	1.06 - 3.61
#Monocyte	0.62	0.57	0.48	0.56	0.30	0.26 - 0.85
#Eosinophil	0.01 L	0.03	0.03	0.03	0.01 L	0.02 - 0.45
#Basophil	0.01 L	0.08	0.00	0.02	0.03	0.02 - 0.00



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Comment	Below	Below	Below	Below		
pH	7.41			7.38	7.35 - 7.45	
Ca++ (measured)	1.53 H			1.45 H	1.12 - 1.23	mmol/L
Ca++ (pH 7.4)	1.54 H			1.44 H	1.12 - 1.23	mmol/L
Na		129 L	130 L	133 L	136 - 148	mmol/L
K		4.7	5.0	5.2 H	3.6 - 5.0	mmol/L
Chloride		97 L	97 L	100	100 - 109	mmol/L
HCO3				14 L	21 - 31	mmol/L
Urea		40.9 H	47.3 H	52.5 H	2.9 - 8.0	mmol/L
Creatinine	→	687 H	763 H	862 H	49 - 82	umol/L
Estimated GFR		5 L	4 L	4 L	>90	unit
Calcium		3.14 H	3.06 H	2.99 H	2.24 - 2.63	mmol/L
Adjusted Calcium	→	3.42 H	3.34 H	3.27 H	2.24 - 2.63	mmol/L
Phosphate		3.15 H	3.18 H	3.08 H	0.88 - 1.45	mmol/L
Total Protein		85		82	67 - 87	g/L
Albumin		28 L	28 L	28 L	39 - 50	g/L
Globulin		57 H		54 H	26 - 40	g/L
Total Bili		18		14	4 - 23	umol/L
ALP	→	117	113	101	47 - 124	U/L
ALT		149 H		99 H	8 - 45	U/L
AST		110 H		55 H	15 - 37	U/L
GGT		89 H		65 H	up to 35	U/L
LDH				132 L	143 - 280	U/L
Urate				878 H	204 - 492	umol/L
Comment :						

GENERAL BIOCHEMISTRY



- M/74
- BPH
- HT
- DM

Collect Date :	12/03/12	12/03/12		
Collect Time :	10:31	14:48		
Arrive Date :	12/03/12	13/03/12		
Arrive Time :	14:53	09:24		
Request No. :	I0028945	I0029123		
Urgency :	--	--		
			Reference	Units
			Range	
IgG	3520 H	--	819 - 1725	mg/dl
IgA	28 L	--	70 - 386	mg/dl
IgM	111	--	55 - 307	mg/dl
B2M	3.32 H	--	< 1.42	ug/ml
SPE	MD	--		
Total protein	84.00	--		g/L
Paraprotein 1	WMK	--		
% Paraprotein 1	WQ	--		%
Paraprotein 2	GK	--		
% Paraprotein 2	28.70	--		%
Concentration 2	24.11	--		g/L
UPE	--	NMD		



Mr K



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- M/74
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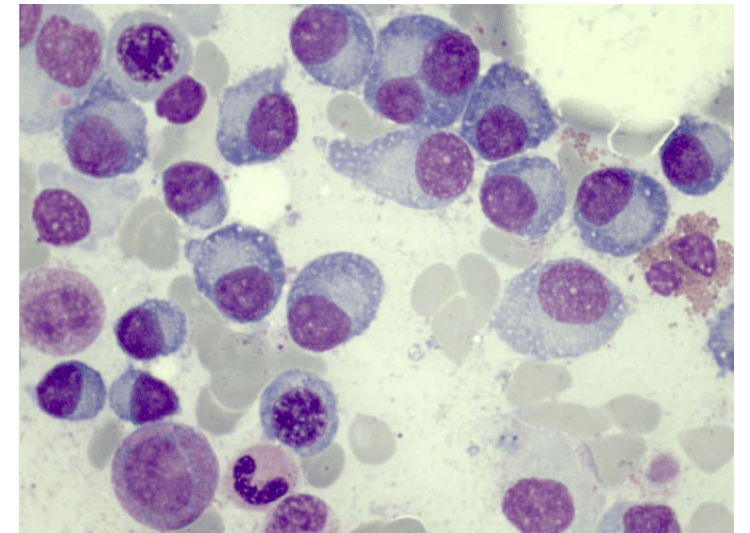


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- Bone marrow examination:
 - Pleomorphic plasma cells accounting for 64% of nucleated cells in the bone marrow
 - Trilineage haematopoiesis is markedly reduced



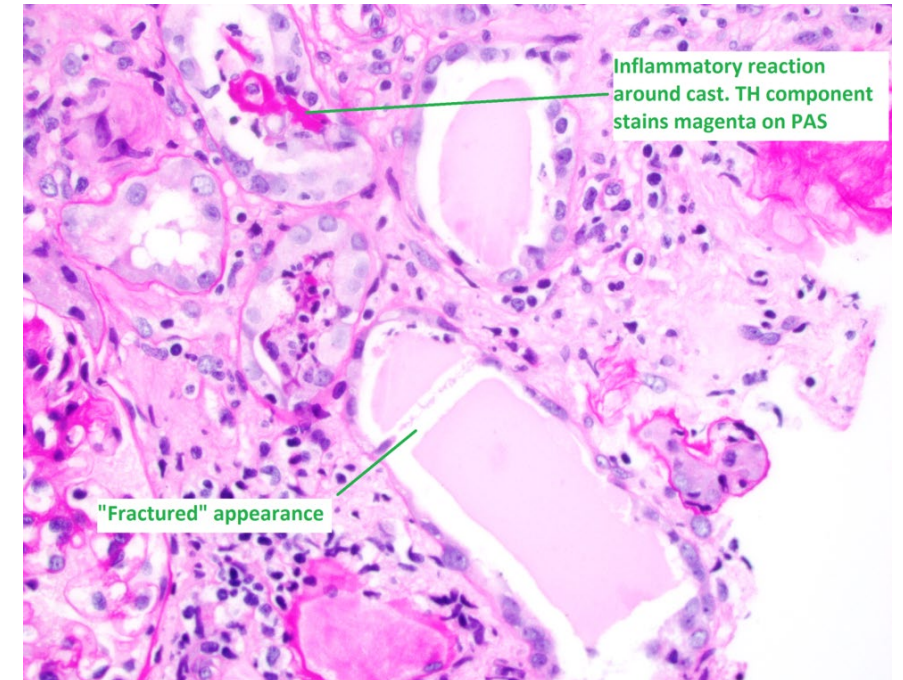


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- Renal Biopsy
 - Mild tubular atrophy and interstitial fibrosis
 - Angulated casts in the lumen of several renal tubules
 - No deposits of immunoglobulin
 - Congo red stain negative



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What is the diagnosis?

Myeloma: epidemiology

- 1% of all malignancy and 10% of haematological malignancies
- Age-standardized incidence: around 5 per 100,000
- Slightly more common in men
- Median age: 66-70 years
- Extremely rare in less than 30 years of age

Pathogenesis

- Multiple myeloma is a neoplastic proliferation of a single clone of terminally differentiated plasma cells

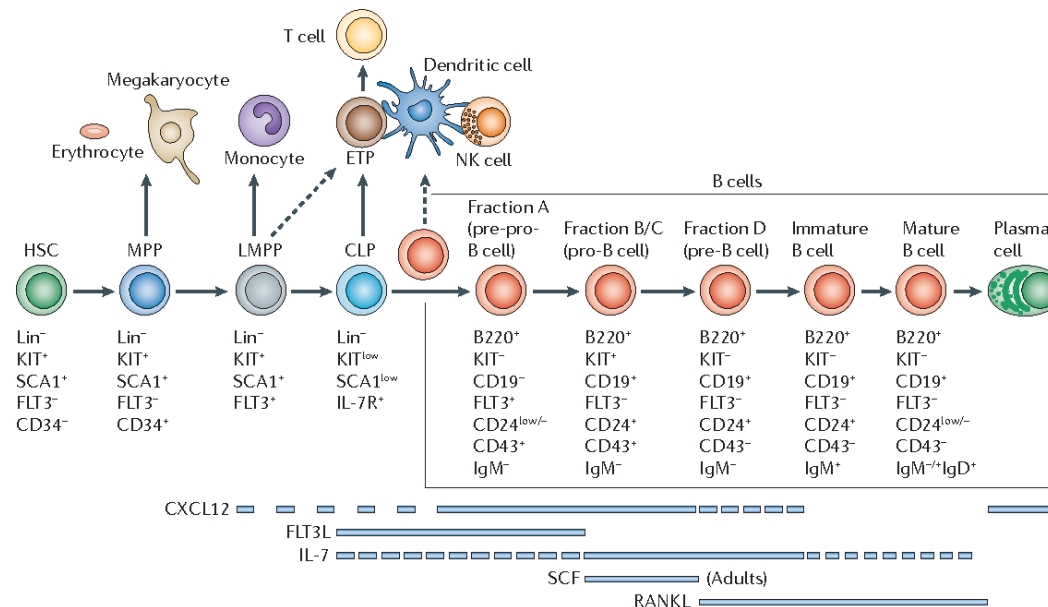


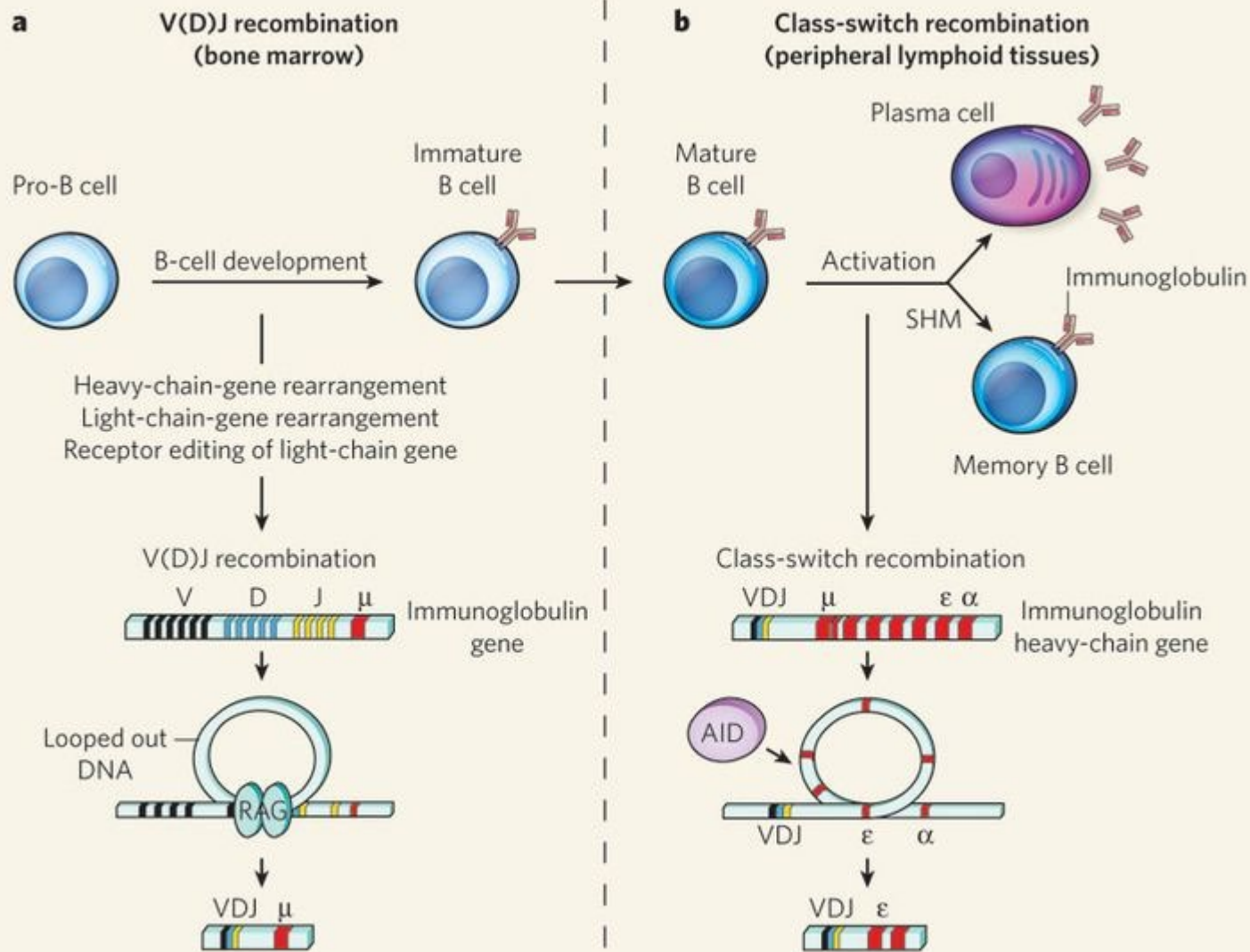
Figure 2 | A pathway for early B cell differentiation and environmental factor requirement in the bone marrow



Pathogenesis

- Clonal plasma cells produced monoclonal immunoglobulin – paraprotein in serum and urine
- Secretion of cytokines causing osteolytic bone lesions and hypercalcaemia

Pathogenesis

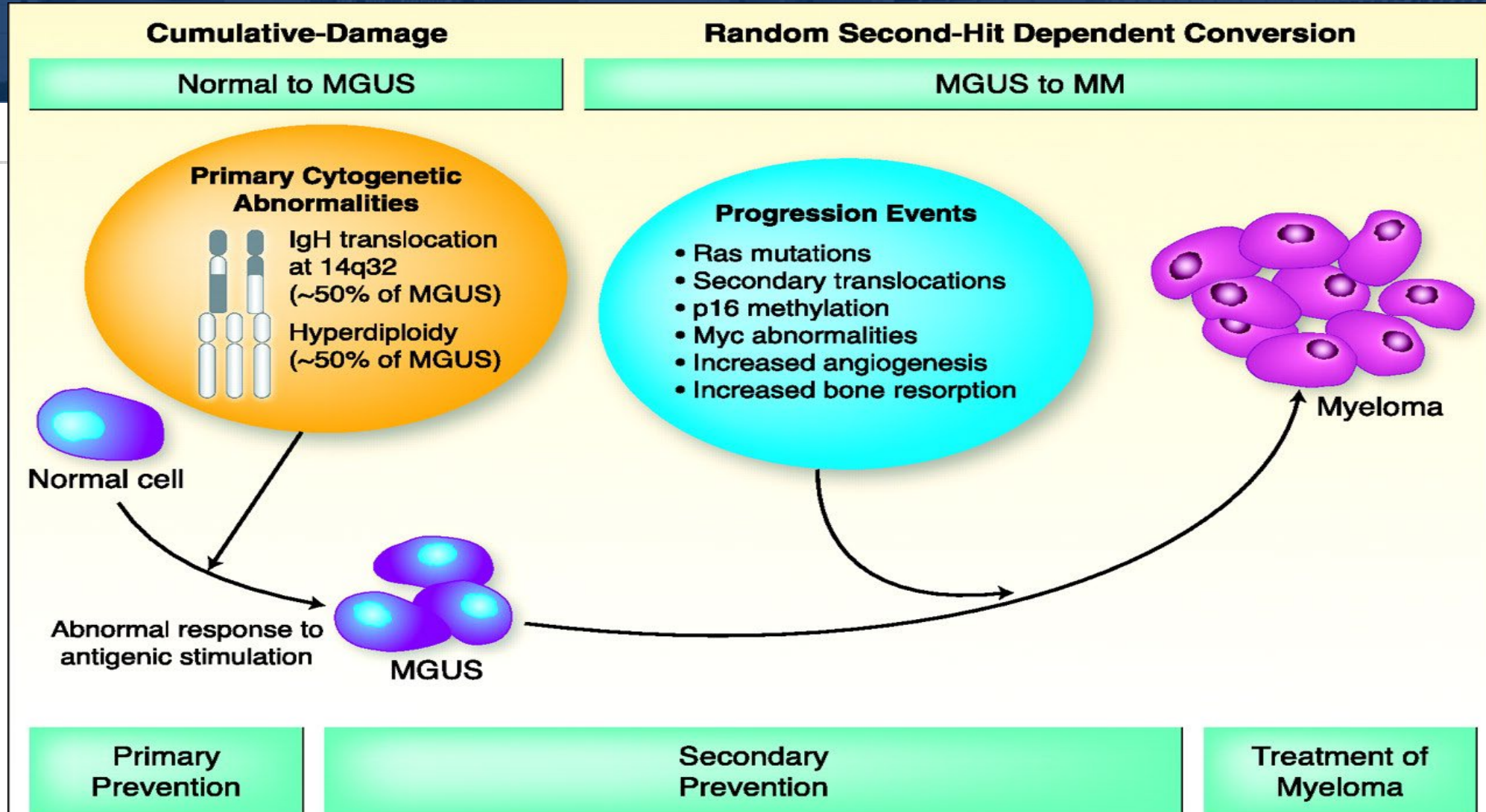


- Normal B cell differentiation
- VDJ rearrangement
 - Somatic hypermutation
 - Class switch recombination

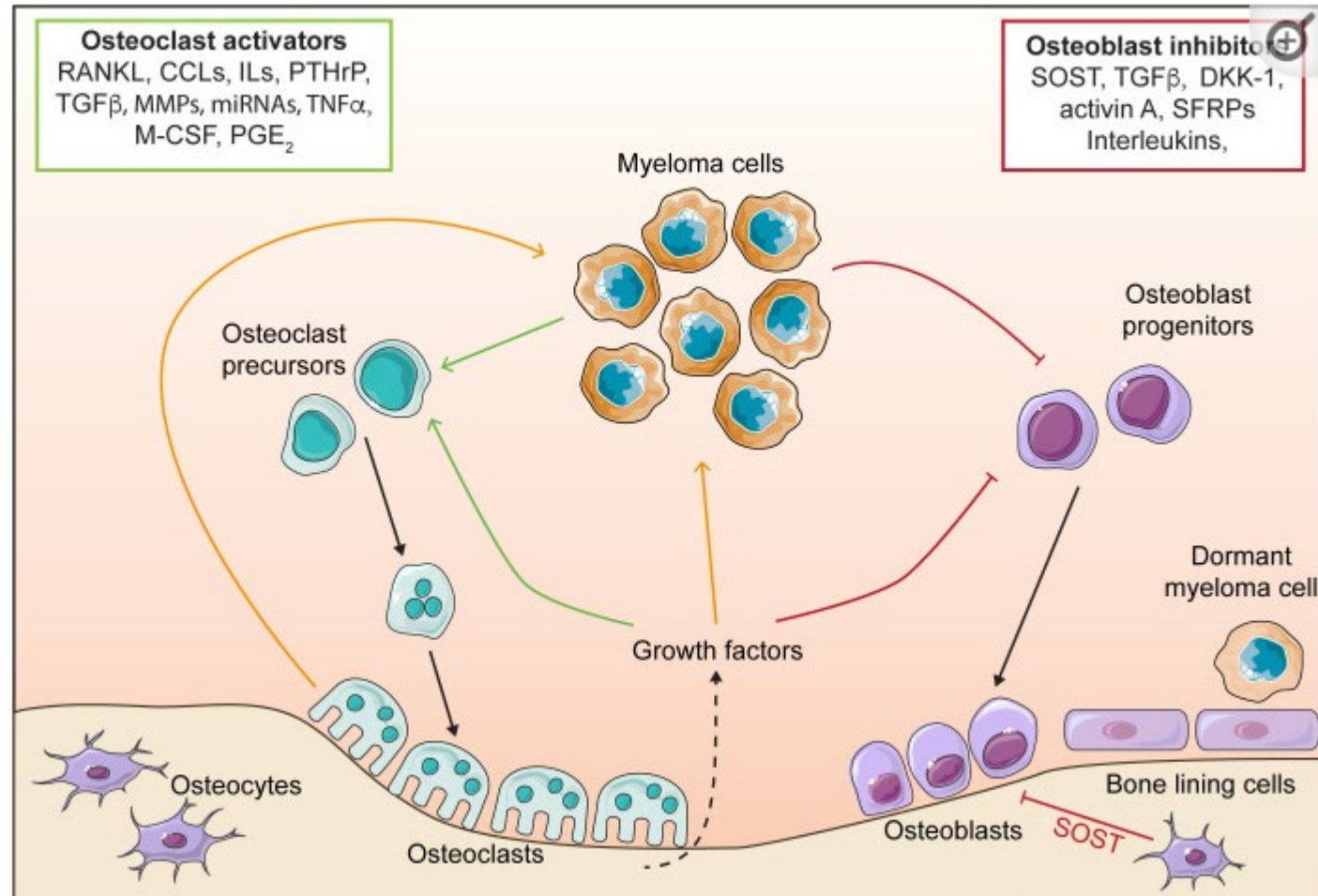
Pathogenesis

- IGH translocations in 40-50% of cases
- 5 recurrent patterns: 11q13(*CCND1*), 4p16(*FGFR/MMSET*), 16q23 (*MAF*), 6p21 (*CCND3*) and 20q11 (*MAFB*)
- Hyperdiploidy with trisomies involving odd chromosomes (3,5,7, 9,11, 15 and 19) is another common karyotypic abnormality

Pathogenesis

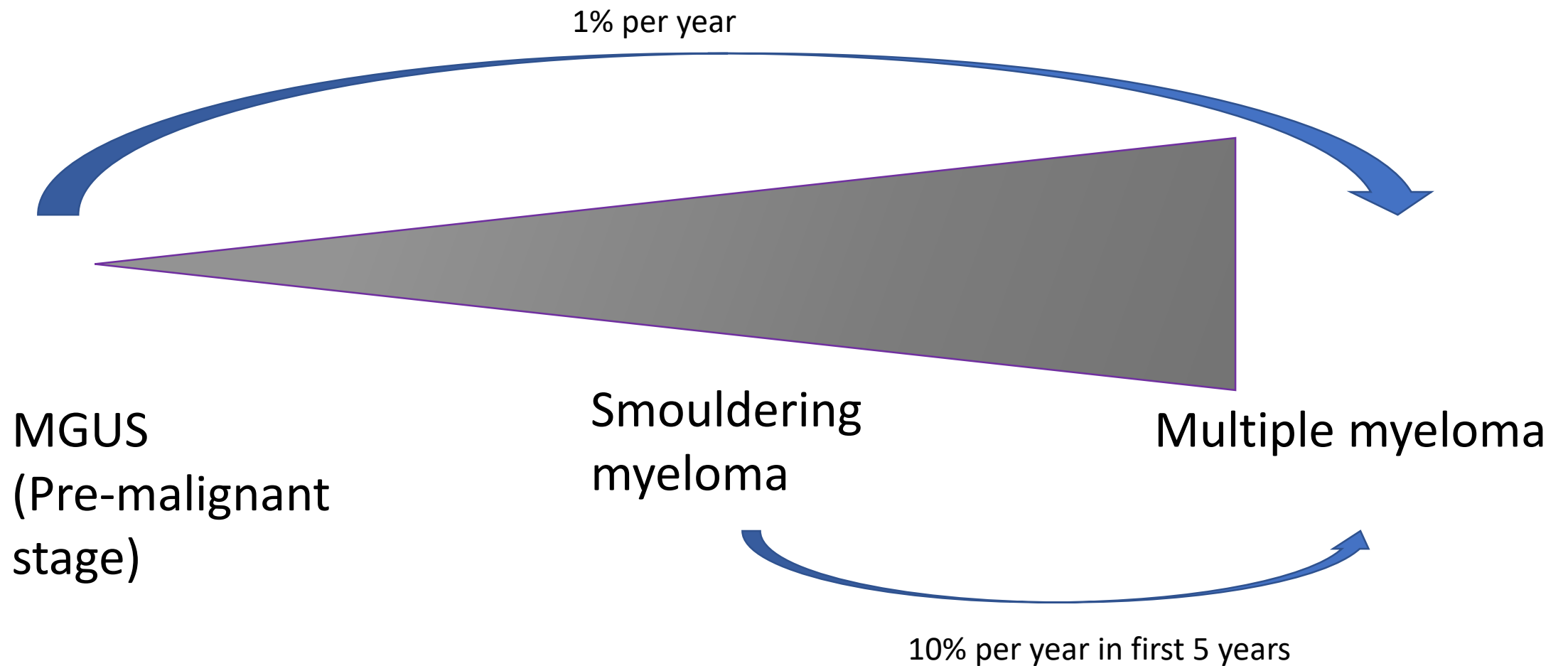


Pathogenesis - microenvironment



Bone marrow microenvironment provides a sanctuary for myeloma cells. Neoplastic plasma cells escape apoptotic pathway and continue to proliferate in bone marrow, and the eventual development of drug resistance.

Development of myeloma



Monoclonal gammopathy of undetermined significance

- ~5% of population above 50 years of age
- Presence of paraprotein of less than 30g/L AND bone marrow plasma cells of less than 10% without any evidence of B cell lymphoproliferative disorder AND absence of organ damage
- AL amyloidosis is a Ddx if patient has suggestive symptoms
- Risk of progression to myeloma
 - Size of paraprotein
 - Type of paraprotein
 - Light chain ratio
 - Rate of rise in paraprotein
 - Bone marrow morphology

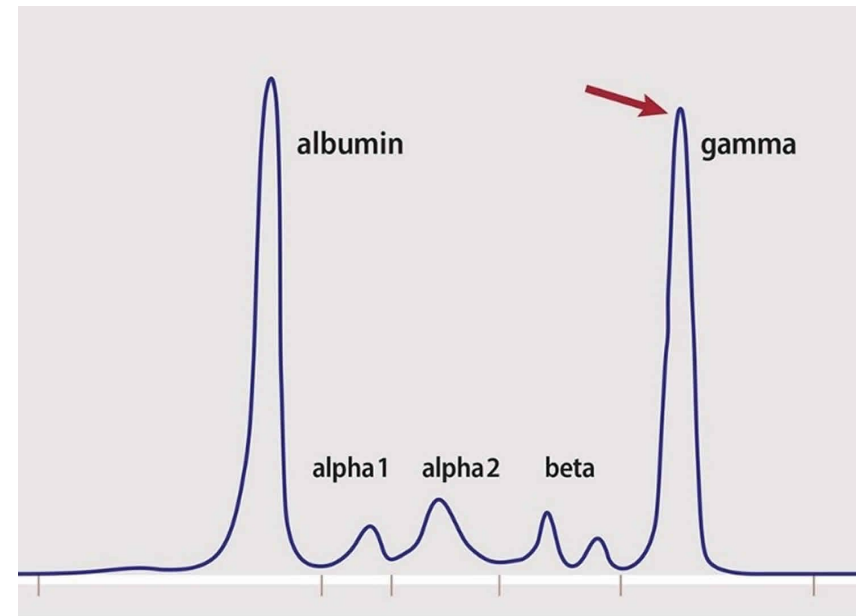
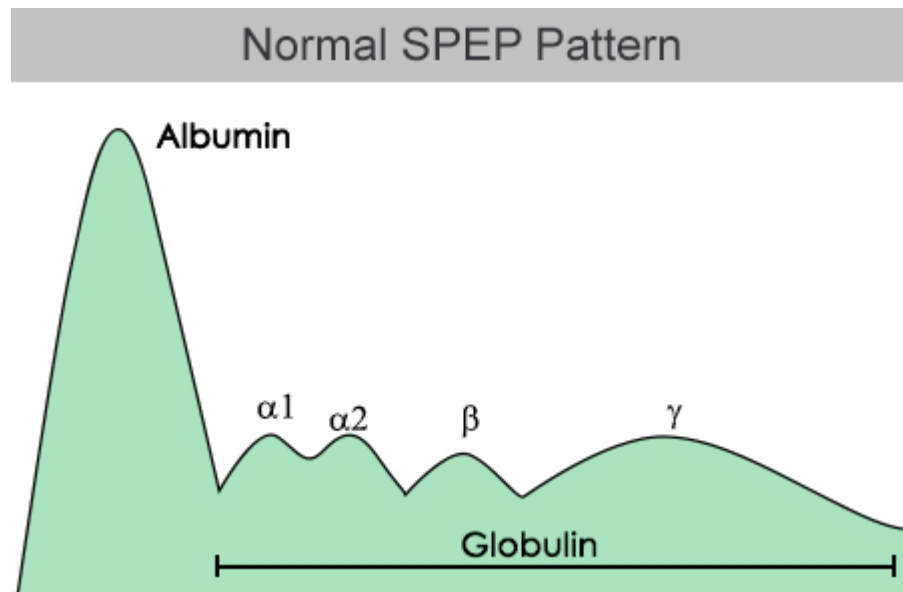


Concept

Defining paraproteins

- Normally our serum immunoglobulin represent the **combined output from millions of different plasma cells**, they are polyclonal
- **Paraproteinaemia** refers to the situation where there is a **monoclonal immunoglobulin band** in the blood secreted by **single clone** of neoplastic cells, usually plasma cells
- The abnormal protein is called **paraprotein** and is detected by **protein electrophoresis of the serum**

Serum protein electrophoresis



Densitometer tracing



Smouldering Myeloma

- Presence of more than 10% clonal plasma cells in myeloma AND paraprotein of >30g/L
- Absence of end organ damage
- Risk of progression to symptomatic myeloma 10% per year in first 5 years then 3% in subsequent 5 years

Diagnosis of multiple myeloma

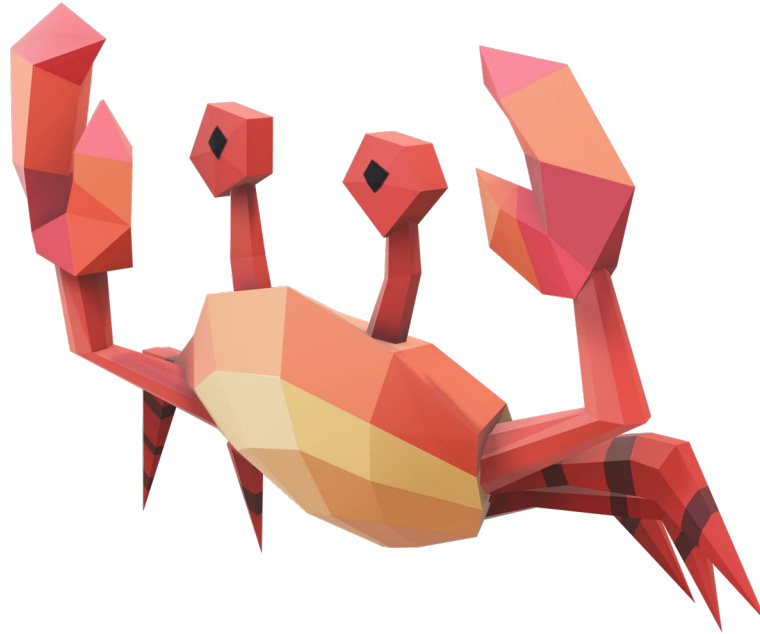
International Myeloma Working Group (IMWG) 2014 diagnostic criteria:

- **Clonal bone marrow plasma cells $\geq 10\%$ or biopsy-proven bony or extramedullary plasmacytoma**
- **One or more** myeloma-defining events
 - Evidence of end organ damage that can be attributed to the underlying plasma cell proliferative disorder
 - **Hyper**Calcaemia: Serum calcium level > 0.25 mmol/L (> 1 mg/dL) higher than the upper limit of normal or > 2.75 mmol/L (> 11 mg/dL)
 - **R**enal insufficiency: (creatinine > 2 mg/dL [> 177 μ mol/L] or creatinine clearance < 40 mL/min)
 - **A**nemia: (hemoglobin < 10 g/dL or hemoglobin > 2 g/dL below the lower limit of normal)
 - **B**one lesions: One or more osteolytic bone lesions on skeletal radiography, CT, or FDG PET-CT
 - Any one or more of the following biomarkers of malignancy
 - Clonal bone marrow plasma cells $\geq 60\%$
 - Involved:uninvolved serum free light chain ratio ≥ 100 with involved free light chain ≥ 100 mg/L
 - One or more (≥ 5 mm) focal lesions on MRI scans

Clinical entities with a paraprotein other than myeloma/MGUS

- Some low grade B cell lymphoma
- Lymphoplasmacytic lymphoma (IgM paraprotein)
- AL amyloidosis
- POEMS / TEMPI syndrome

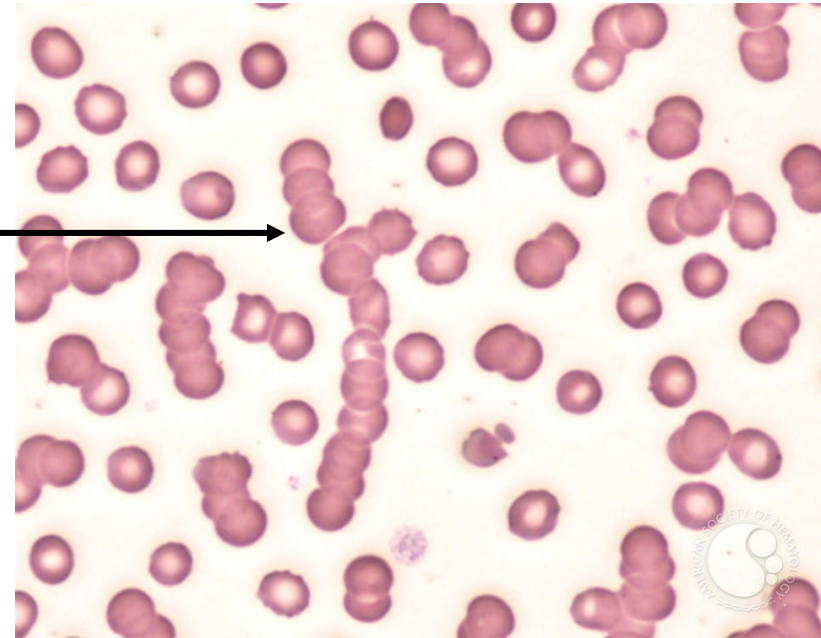
Complications of Myeloma



Myeloma related complications: Anaemia

Why myeloma patients are anemic?

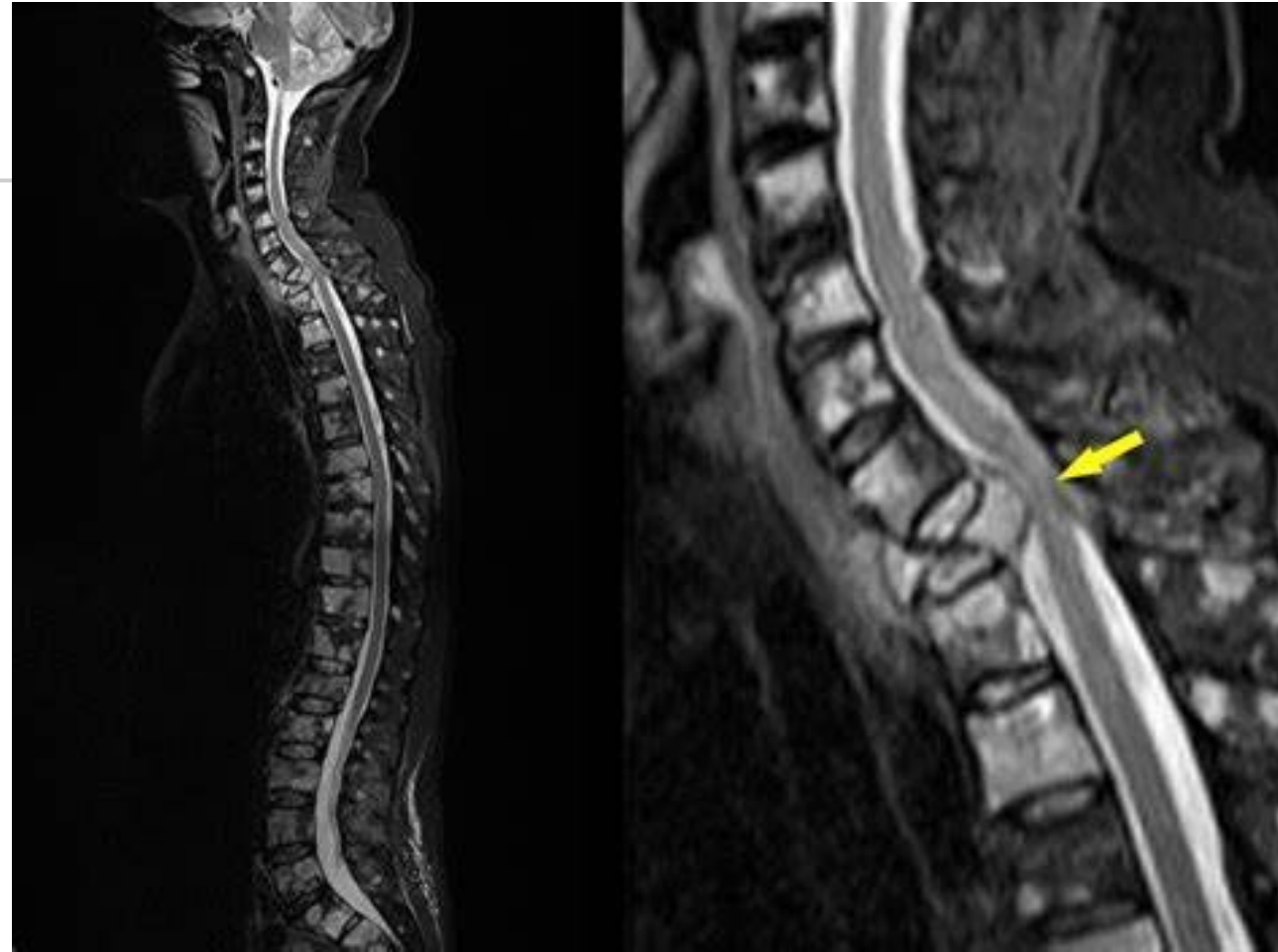
- Marrow replacement by plasma cells
- Reduced erythropoietin
- Dilution (by paraprotein)
- Rouleaux of red cells



Myeloma related complications: Bone/Calcium

- **Markedly increased bone resorption** secondary to increase in osteoclastic activity, caused by secretion of potent cytokines including receptor activator of nuclear factor κ -B ligand (RANKL), macrophage inflammatory factor and tumour necrosis factors
- **Direct myeloma deposit** causing radiological appearance of punch-out lesion on skull
- Characteristically, **osteoblastic activity is NOT increased** and therefore in typical myeloma bone disease, alkaline phosphatase (**ALP**, a marker of osteoblastic activity) is **NOT elevated**
- It explains the hypercalcaemia (bone resorption and leakage of calcium into systemic circulation) and bone lesions (widespread decrease in bone density, punch-out lesions on skull, propensity to pathological fracture)

Myeloma related complications: Bon

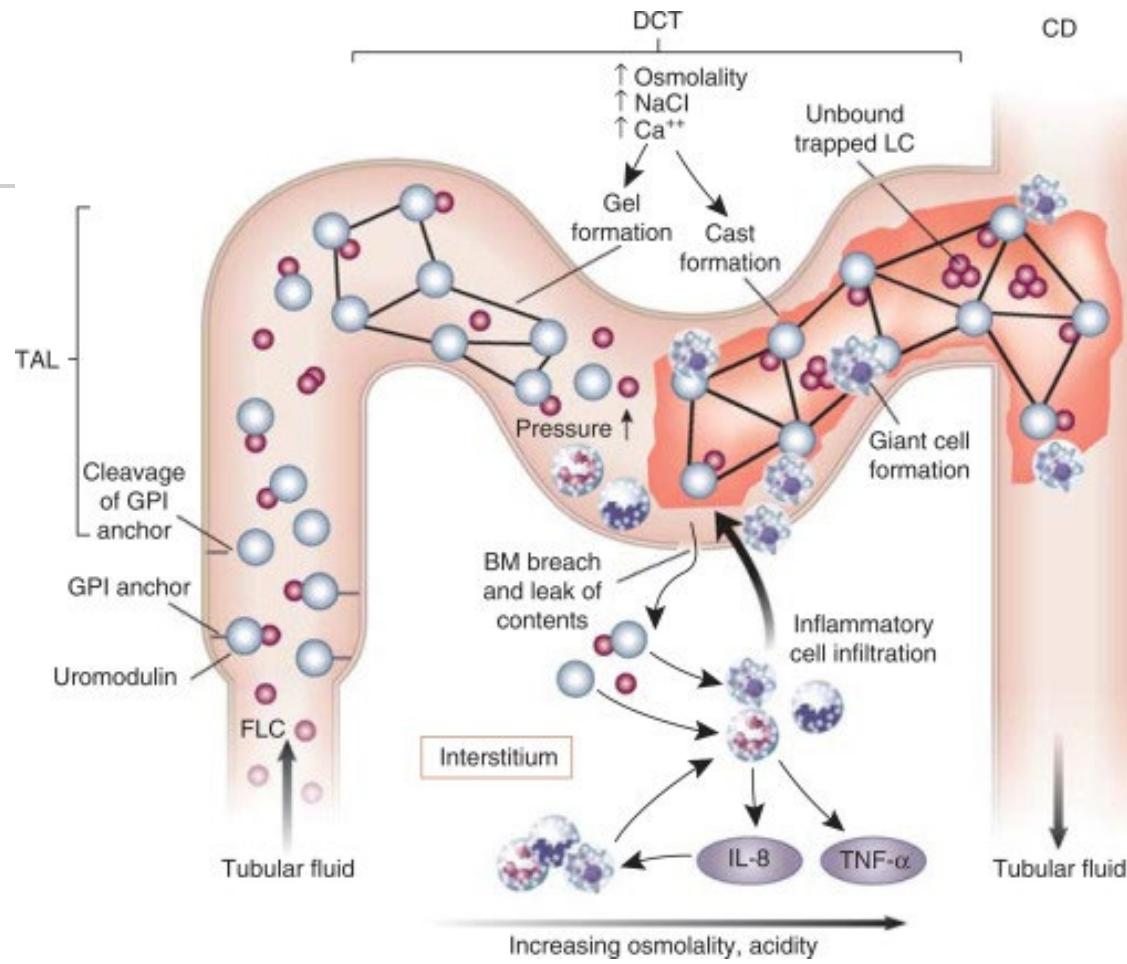


Extramedullary plasmacytoma or fracture of vertebral body could lead to spinal cord compression

Myeloma related complications: kidney

- At least 50% of patients with newly diagnosed myeloma have renal involvement of various degree
- **Cast nephropathy** is the predominant cause of kidney injury, which is also known as myeloma kidney
- Other mechanisms of kidney injury includes: **hypercalcaemia, dehydration, amyloidosis and direct myeloma deposition** (the last two are rare)

Cast nephropathy



- Excessive clonal free light chains which are filtered freely through the glomeruli, and exceeded the re-absorption by proximal tubular cells
-
- Combined with uromodulin to form waxy cast
-
- Obstruction of distal renal tubules, causing tubulointerstitial nephritis as the tubules rupture.

Work-up for patients with paraprotein

- History and Physical Examination
- Complete blood count
- Renal function and calcium, albumin, ALP, and LDH
- Serum protein electrophoresis +/- immunofixation
- Urine for protein
- Urine protein electrophoresis
- Serum free light chain level and ratio
- Serum beta 2 immunoglobulin
- Bone marrow aspirate and trephine. FISH analysis
- (Skeletal Survey), Whole body PET-CT or whole body MRI or whole body low dose CT
- Dental assessment

Staging and prognosis of myeloma

- Durie Salmon Staging

Stage I

Low cell mass: $<0.6 \times 10^{12}$ cell/ M^2 : ALL of the following:

Hb >10 g/dL

Serum IgG <5000 mg/dL

Serum IgA <3000 mg/dL

Normal serum calcium

Urine monoclonal protein excretion <4 g/D

No generalized lytic bone lesions

Stage II

Intermediate cell mass: neither stage I nor stage III

Stage III

High cell mass: $>1.2 \times 10^{12}$ cells/ m^2 :ONE or more of the following:

Hb <8.5 g/dL

Serum IgG >7000 mg/dL

Serum IgA >5000 mg/dL

Serum calcium >3 mmol/L

Urine monoclonal protein excretion >12 g/D

Advanced lytic bone lesion

Stage IIIA or B

A. Serum creatinine <177 umol/L

B. Serum creatinine ≥ 177 umol/L

Staging and prognosis of myeloma

Revised International Staging System (R-ISS):

5-year Overall survival

Stage I	82%
Stage II	62%
Stage III	40%

Table 1. Standard Risk Factors for MM and the R-ISS

Prognostic Factor	Criteria
ISS stage	
I	Serum β_2 -microglobulin < 3.5 mg/L, serum albumin $\geq$ 3.5 g/dL
II	Not ISS stage I or III
III	Serum β_2 -microglobulin $\geq$ 5.5 mg/L
CA by iFISH	
High risk	Presence of del(17p) and/or translocation t(4;14) and/or translocation t(14;16)
Standard risk	No high-risk CA
LDH	
Normal	Serum LDH < the upper limit of normal
High	Serum LDH > the upper limit of normal
A new model for risk stratification for MM	
R-ISS stage	
I	ISS stage I and standard-risk CA by iFISH and normal LDH
II	Not R-ISS stage I or III
III	ISS stage III and either high-risk CA by iFISH or high LDH

Abbreviations: CA, chromosomal abnormalities; iFISH, interphase fluorescent in situ hybridization; ISS, International Staging System; LDH, lactate dehydrogenase; MM, multiple myeloma; R-ISS, revised International Staging System.

Staging and Prognosis

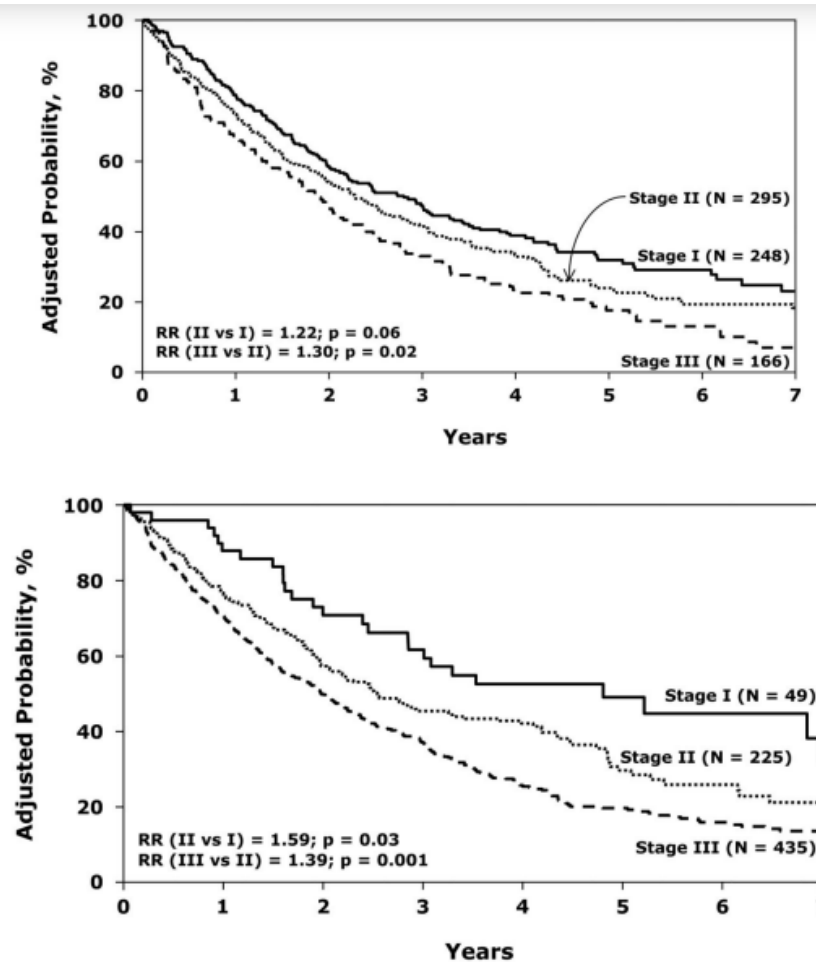
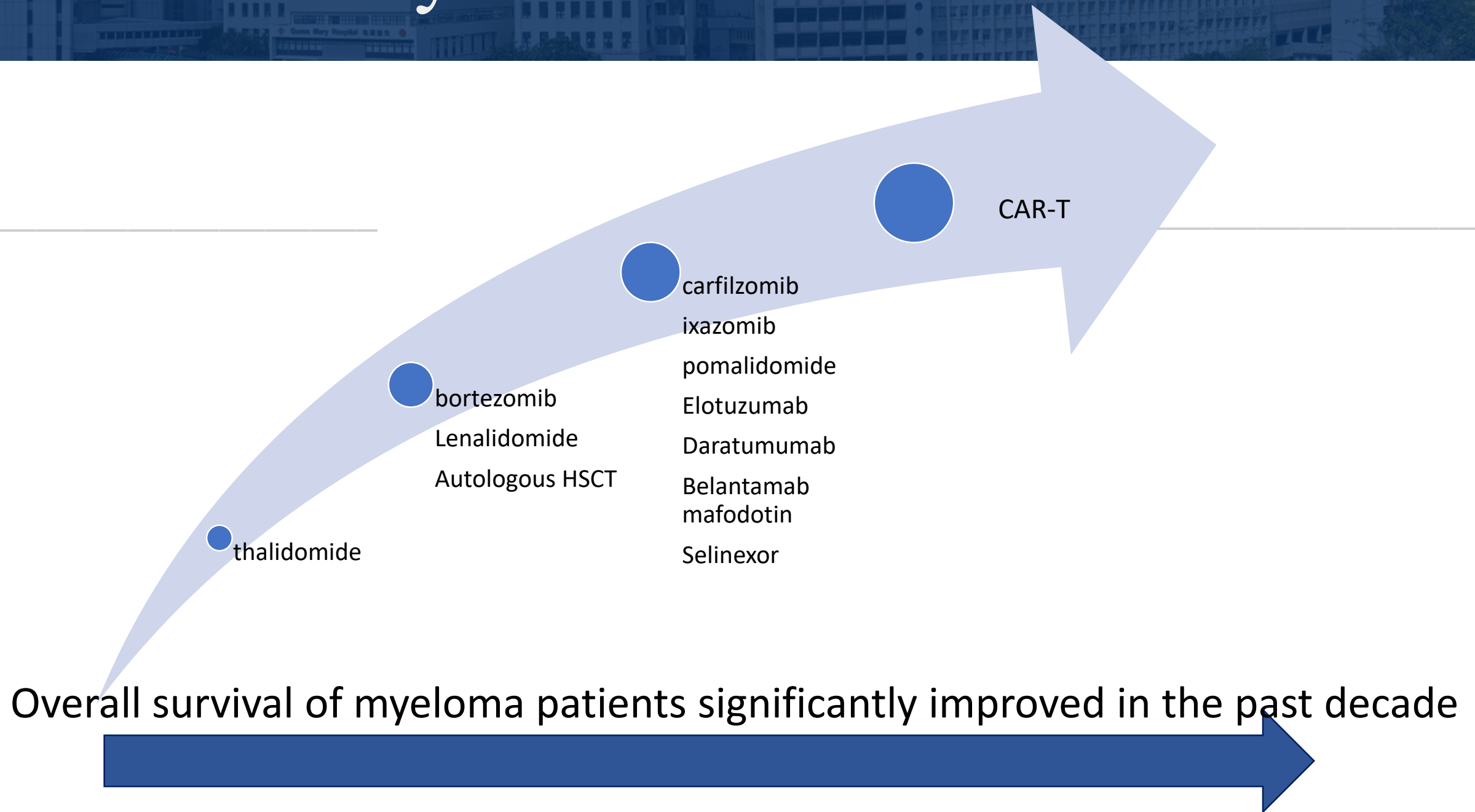


Figure 1. Adjusted probabilities of progression-free survival after upfront autologous stem cell transplantation for Multiple Myeloma, by International Staging System (A) and Durie-Salmon System (B).

Prognosis: FISH

Standard Risk (60%)	Intermediate Risk (20%)	High Risk (20%)
t(11;14)	t(4;14)	Del17p
t(6;14)	Del13	t(14;16)
Hyperdiploid	Hypodiploid	t(14;20)
		(1q gain)
		(P53 mutation)
All others		GEP high risk signature
OS 8-10 years	OS 4-5 years	OS 3 years

Myeloma treatment



Myeloma treatment

Figure 1A

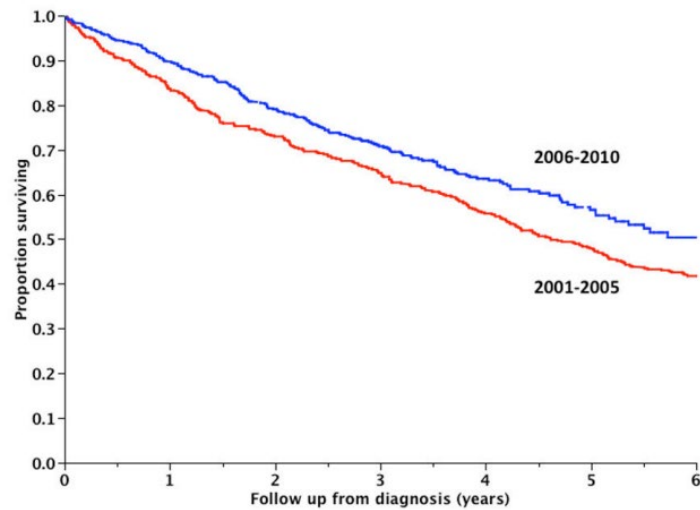


Figure 1B

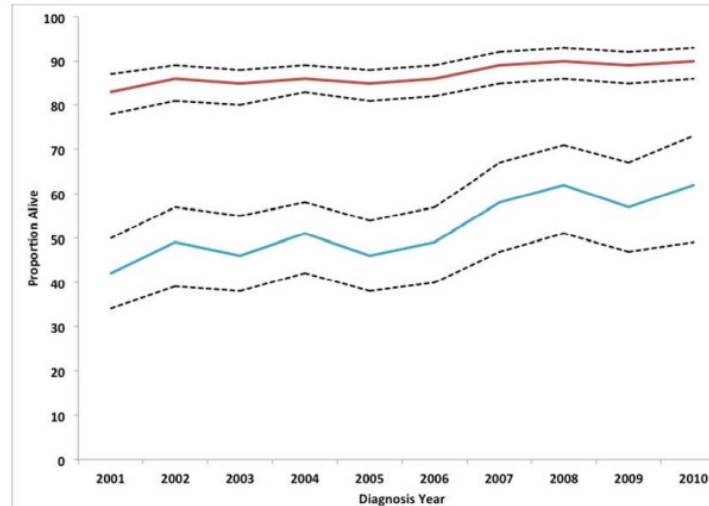
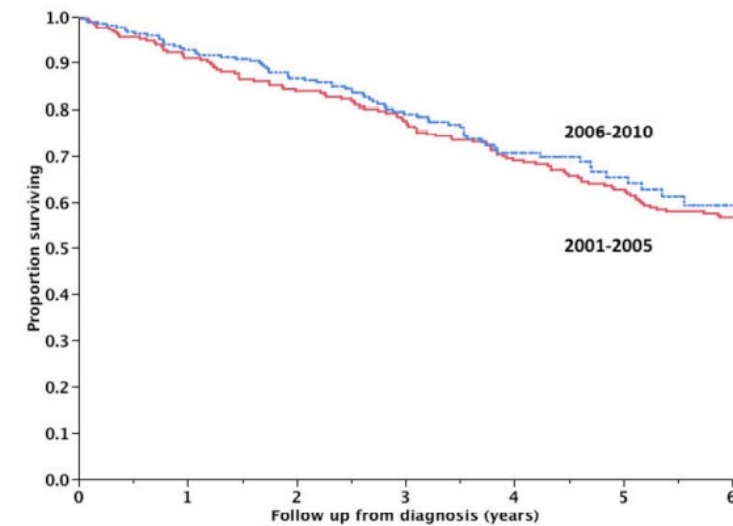
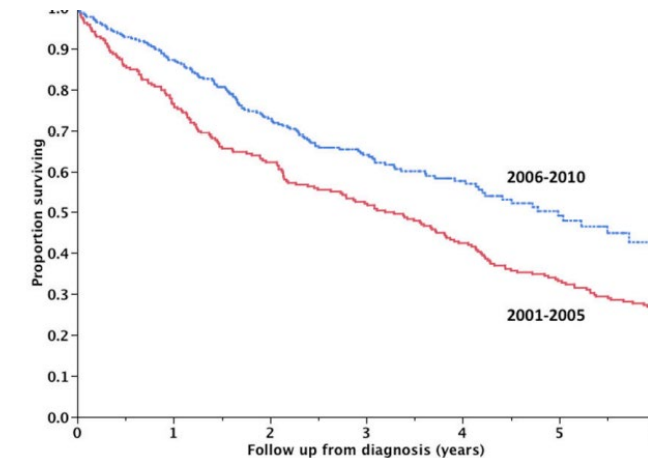


Figure 1C



Panel A shows the overall survival comparison between patients diagnosed during January 2001–December 2005 and those diagnosed during January 2006 and December 2010. Panel B shows the trends in the 1 and 5 year overall survival estimates between January 2001 and December 2010 with patients grouped by the year of diagnosis. The dotted lines represent the 95% confidence intervals. Panel C shows the overall survival comparison between patients diagnosed during 2001–2005 and those diagnosed during 2006–2010 limited to patients 65 years or younger. Panel D shows the overall survival comparison between patients diagnosed during 2001–2005 and those diagnosed during 2006–2010 limited to patients older than 65 years.

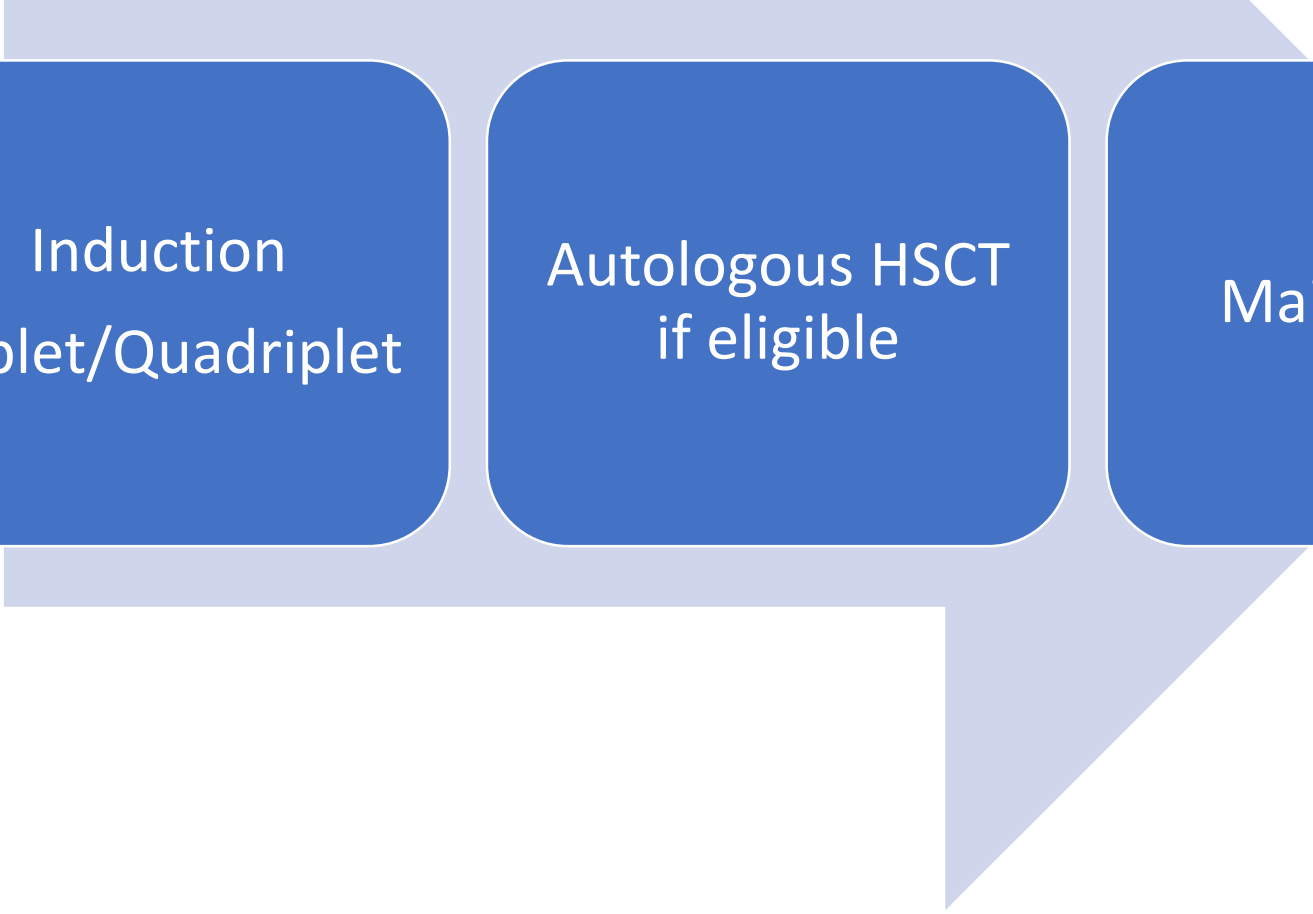


Treatment planning

Important considerations:

- Eligibility for autologous haematopoietic stem cell transplant (ASCT)
- Presence of high risk features:
 - i. Plasma cell leukaemia
 - ii. Markedly elevated LDH
 - iii. FISH abnormalities: t(4;14) ; t(14;16) ; t(14;20) ; del(17p) ; Gain 1q
- The need of adjunctive treatment including radiotherapy and anti-resorptives

Treatment planning for transplant eligible patients

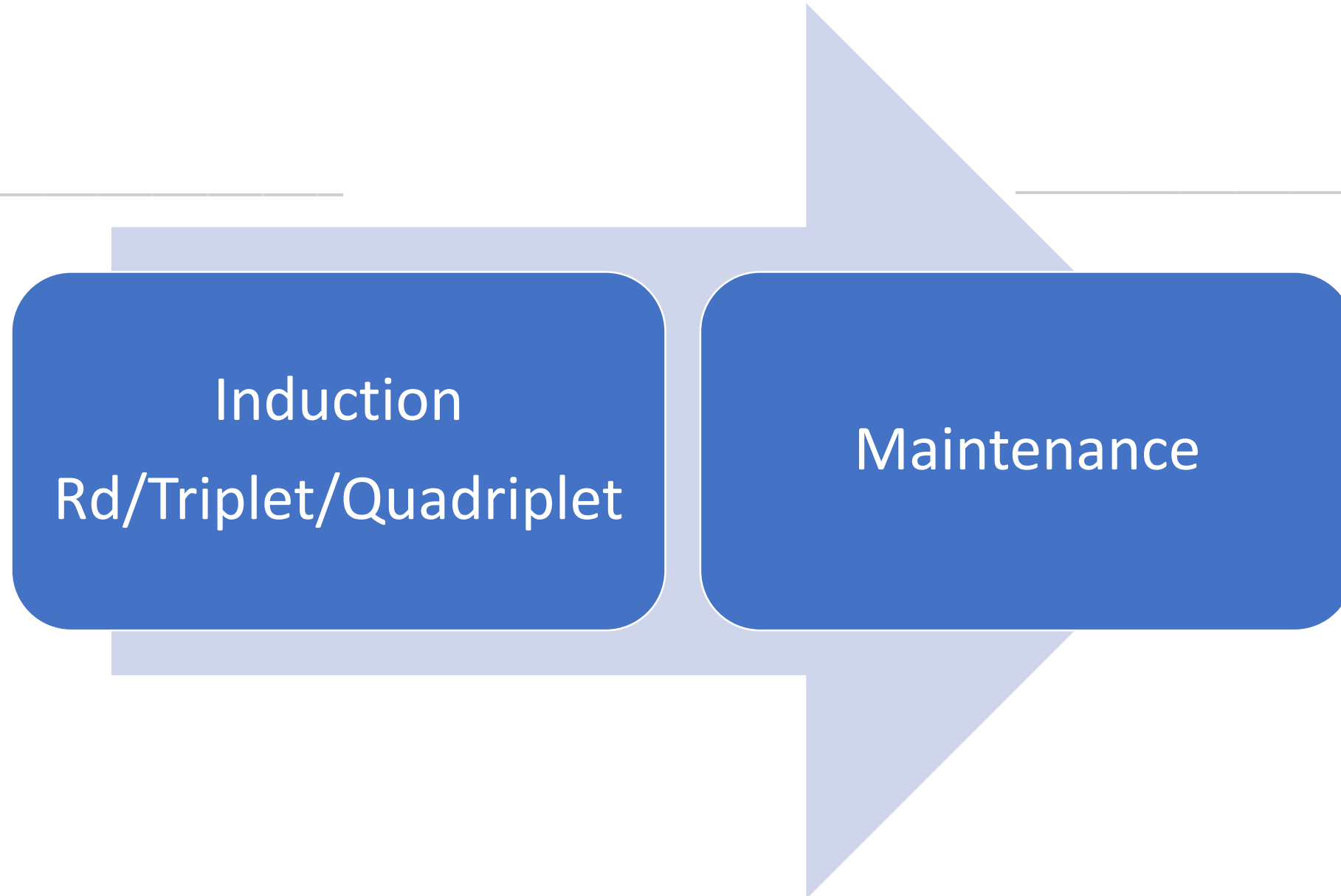


Induction
Triplet/Quadriplet

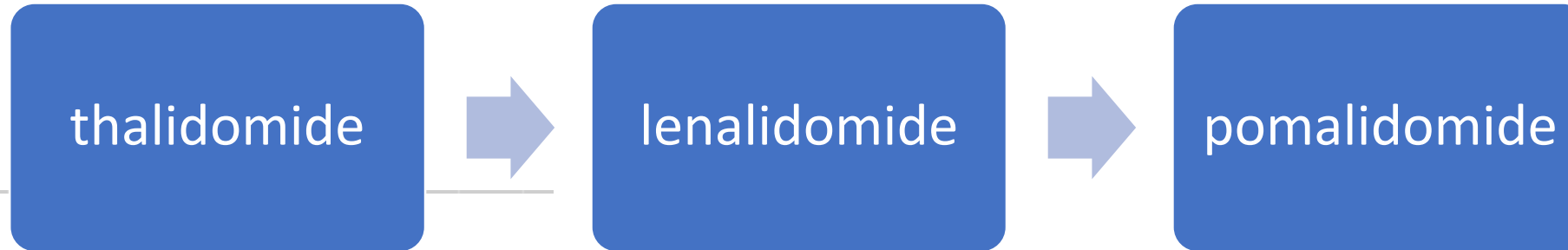
Autologous HSCT
if eligible

Maintenance

Treatment planning for transplant ineligible patients



IMiDs Immunomodulatory Agents



- bind to cereblon and activate cereblon E3 ligase activity
- rapid ubiquitination and degradation of two specific B cell transcription factors, Ikaros family zinc finger proteins Ikaros (IKZF 1) and Aiolos (IKZF3)
- Anti-angiogenesis and immunomodulatory effect
- increases risk of thromboembolism

Proteasome Inhibitor



- Inhibition of 26S proteasome
- Prevent proteolysis of tagged protein
- Cytotoxic to myeloma cells by affecting multiple signaling cascade within cells

Treatment planning

Novel agents:

- Monoclonal antibodies: daratumumab, elotuzumab, belantamab mafodotin
- Nuclear cytoplasmic transport receptor inhibitor: Selinexor
- BCL-2 inhibitors: venetoclax
- CAR-T cell therapy

After today's lecture

You should be able

1. to recognize clinical features of multiple myeloma
2. to understand the clinical spectrum of plasma cell dyscrasia
3. to tell the potential complications of myeloma and their underlying mechanism
4. to list treatment options of myeloma



Thank you!
